Anaphora and Coreference Resolution using spaCy

File: anaphora_coreference.py [newhorizon...epoint.com]

Objective

This program demonstrates a simple **Anaphora and Coreference Resolution** system.

Anaphora Resolution

Identifies the noun referred to by a pronoun.

Example:

```
Rahul is a student.
He studies computer science.
```

Here:

```
He → Rahul
```

Coreference Resolution

Determines which words in a text refer to the same entity.

Example:

```
Rahul likes Python.
He wants to become a software engineer.
```

Both **Rahul** and **He** refer to the same person.

The program uses **spaCy** to identify named entities and pronouns and then links pronouns with the most recently mentioned person. [newhorizon...epoint.com]

Line 1

```
import spacy
```

Explanation

Imports the **spaCy** library.

spaCy provides:

- Tokenization
- POS Tagging
- Named Entity Recognition (NER)

- Dependency Parsing

The program uses spaCy to identify:

- Person names
- Pronouns

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Load the NLP Model

Line 4

```
1  nlp = spacy.load("en_core_web_sm")
```

Explanation

Loads spaCy's English language model.

The model provides:

- Tokenizer
- POS Tagger
- NER Model

This allows the program to identify entities such as:

```
1  PERSON
2  ORG
3  GPE
4  DATE
```

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Sample Text

Lines 7-12

```
1  text = """
2  Rahul is a student.
3  He studies computer science.
4  Rahul likes Python.
5  He wants to become a software engineer.
```

```
"""
```

Explanation

Input document containing:

- A person name: Rahul
- Pronouns: He

The task is to determine:

```
He → Rahul
```

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Process the Text

Line 15

```
doc = nlp(text)
```

Explanation

Processes the text through spaCy.

spaCy performs:

```
Text
 ↓
Tokenization
 ↓
POS Tagging
 ↓
Named Entity Recognition
 ↓
Doc Object
```

The result is stored in:

```
doc
```

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Display Title

Line 17

```
1  print("ANAPHORA AND COREFERENCE RESOLUTION")
```

Explanation

Displays the experiment title.

Output:

```
1  ANAPHORA AND COREFERENCE RESOLUTION
```

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Line 18

```
1  print("----------------------------------")
```

Explanation

Prints a separator line.

Output:

```
1  ----------------------------------
```

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Create Entity List

Line 21

```
1  entities = []
```

Explanation

Creates an empty list.

This list stores person entities found in the text.

Example:

```
1  ['Rahul']
```

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Extract PERSON Entities

Line 22

```
1  for ent in doc.ents:
```

Explanation

Loops through all named entities detected by spaCy.

Examples:

```
1  Rahul
2  Python
```

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Line 23

```
1  if ent.label_ == "PERSON":
```

Explanation

Checks whether the entity type is:

```
1  PERSON
```

Only person names are collected.

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Line 24

```
1  entities.append(ent.text)
```

Explanation

Adds the person's name to the list.

Result:

```
1  ['Rahul', 'Rahul']
```

Since Rahul appears twice in the text.

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Display Named Entities

Line 26

```
print("\nNamed Entities:")
```

Explanation

Prints a heading.

Output:

```
Named Entities:
```

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Lines 27-28

```
for entity in entities:
    print(entity)
```

Explanation

Displays all identified person names.

Output:

```
Rahul
Rahul
```

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Anaphora Resolution Section

Line 31

```
print("\nPronouns and their Antecedents:")
```

Explanation

Displays the title for pronoun resolution results.

Output:

```
Pronouns and their Antecedents:
```

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Initialize Antecedent

Line 32

```
antecedent = None
```

Explanation

Creates a variable to store the most recent person name.

Initially:

```
antecedent = None
```

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Process Every Token

Line 34

```
for token in doc:
```

Explanation

Loops through every word in the document.

Tokens include:

```
Rahul
is
a
student
.
He
studies
computer
science
 ...
```

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Identify Person Names

Line 36

```
1  if token.ent_type_ == "PERSON":
```

Explanation

Checks whether the current token belongs to a PERSON entity.

Example:

```
1  Rahul
```

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Line 37

```
1  antecedent = token.text
```

Explanation

Stores the latest person name.

Example:

```
1  antecedent = "Rahul"
```

This name becomes the reference for upcoming pronouns.

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Identify Pronouns

Line 40

```
1  if token.pos_ == "PRON":
```

Explanation

Checks whether the current word is a pronoun.

Examples:

```
1  He
2  She
3  They
4  His
```

```
5  Her
6  Their
```

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Male Pronouns

Line 41

```
1  if token.text.lower() in ["he", "him", "his"]:
```

Explanation

Checks for male pronouns.

Examples:

```
1  he
2  him
3  his
```

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Line 42

```
1  print(token.text, "->", antecedent)
```

Explanation

Prints the connection between pronoun and antecedent.

Example:

```
1  He -> Rahul
```

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Female Pronouns

Lines 43-44

```
1  elif token.text.lower() in ["she", "her", "hers"]:
```

Explanation

Checks for female pronouns.

Examples:

```
she
her
hers
```

Output Example:

```
She -> Mary
```

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Plural Pronouns

Lines 45-46

```
elif token.text.lower() in ["they", "them", "their"]:
```

Explanation

Handles plural pronouns.

Examples:

```
they
them
their
```

Output Example:

```
They -> Students
```

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Completion Message

Line 48

```
print("\nCoreference Resolution Completed.")
```

Explanation

Displays the completion message.

Output:

```
Coreference Resolution Completed.
```

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Dry Run

Input Text

```
Rahul is a student.
He studies computer science.
Rahul likes Python.
He wants to become a software engineer.
```

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Step 1: Named Entity Recognition

spaCy identifies:

```
Rahul → PERSON
Rahul → PERSON
```

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Step 2: Store Antecedent

```
antecedent = Rahul
```

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Step 3: Pronoun Resolution

First occurrence:

```
He → Rahul
```

Second occurrence:

```
He → Rahul
```

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Final Output

```
Named Entities:
Rahul
Rahul

Pronouns and their Antecedents:
He -> Rahul
He -> Rahul

Coreference Resolution Completed.
```

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Key Concepts

Term	Meaning
Entity	Real-world object/person identified by NER
Pronoun	Word that replaces a noun
Antecedent	Noun referred to by a pronoun
Anaphora Resolution	Finding the antecedent of a pronoun
Coreference Resolution	Finding all words that refer to the same entity

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Viva Questions

1. What is Anaphora Resolution?
2. What is Coreference Resolution?
3. What is an antecedent?
4. What is a pronoun?
5. How does spaCy identify PERSON entities?
6. What does token.ent_type_ return?
7. What does token.pos_ represent?
8. What is the difference between NER and Coreference Resolution?
9. Why is Coreference Resolution important in NLP?
10. What are the limitations of this rule-based approach?

Result

The program successfully implements a simple **Anaphora and Coreference Resolution** system using spaCy, identifies person names and pronouns, and maps pronouns to their corresponding antecedents to determine references within the text.
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